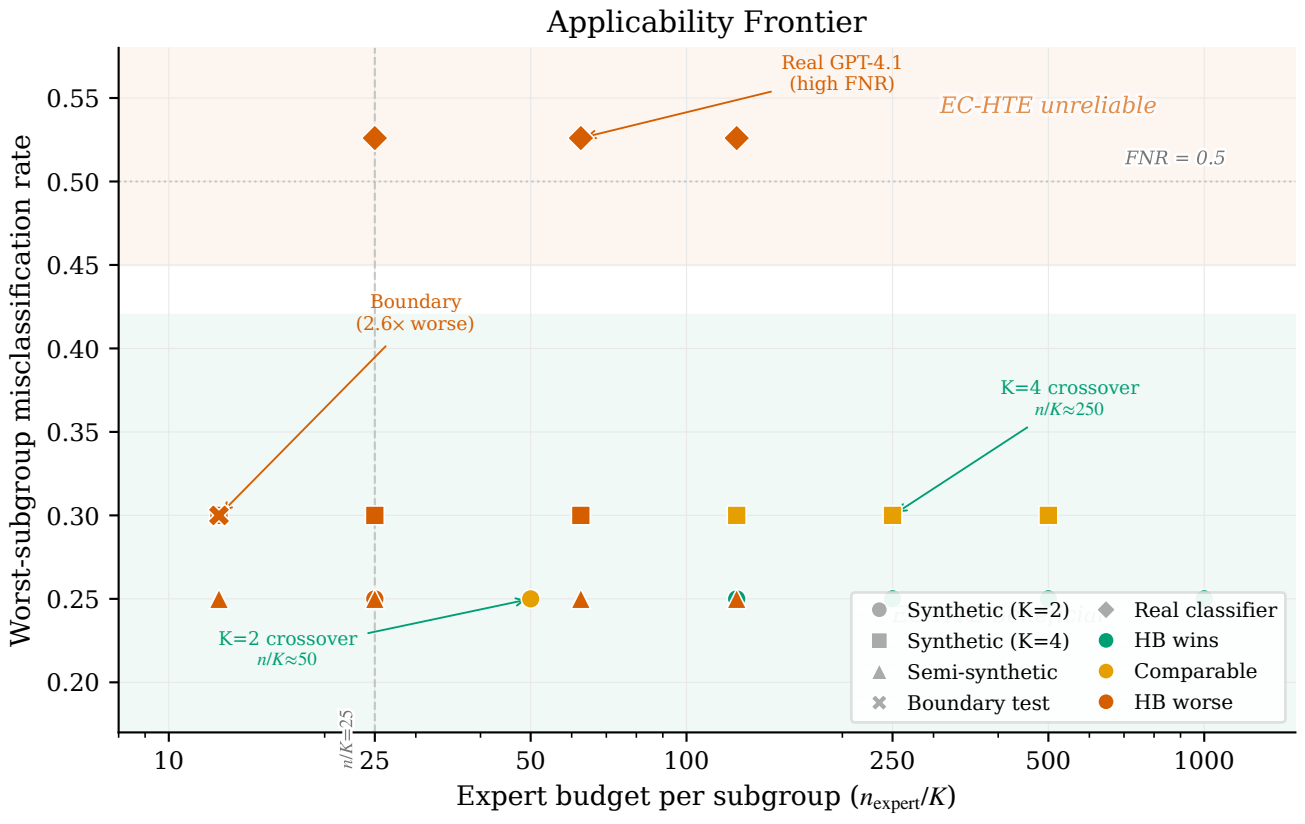
Applicability Frontier
Real GPT-4.1
(high FNR)
EC-HTE unreliable
FNR = 0.5
Boundary
(2.6× worse)
K=4 crossover
n/K≈250
K=2 crossover
n/K≈50
n/K=25
Synthetic (K=2)
Synthetic (K=4)
Semi-synthetic
Boundary test
Real classifier
HB wins
Comparable
HB worse
Worst-subgroup misclassification rate
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0.20
10
25
50
100
250
500
1000
Expert budget per subgroup (n_{expert}/K)